

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \sum_{k=l}^{l+n} \delta_{k,k+1}^2 &\leq 4e_l^2 + 2^6 \varepsilon^{-1} \Lambda_6^2 \mu_l^2 + \left(\frac{1}{2} \varepsilon C_{\text{REL}}^{-2} - 4\right) e_{l+n+1}^2 + \left(\frac{\varepsilon}{4} - 2^6 \varepsilon^{-1} \Lambda_6^2\right) \mu_{l+n+1}^2 \\ &\quad + \frac{\varepsilon}{2} \sum_{k=l}^{l+n} \mu_k^2 + \frac{\varepsilon}{2} C_{\text{REL}}^{-2} \sum_{k=l}^{l+n-1} e_{k+1}^2. \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathbf{W}_p(\nu(t_0), \nu(t_1)) &= \mathbf{W}_p(\nu(t_0), \gamma + \Delta \pi') \\ &\leq (1 - \Delta) \mathbf{W}_p(\nu(t_0), (1 - \Delta)^{-1} \gamma) + \underbrace{\Delta \mathbf{W}_p(\nu(t_0), \pi')}_{=O(\Delta)}. \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} I_2 &= |\vec{\Gamma} \cdot \vec{\Gamma}|^2 = \sum_{n=1}^N |\Gamma^{(n)}|^2, \\ I_4 &= -\frac{1}{4} \det \left[\begin{pmatrix} \vec{\Gamma} \\ \vec{\Gamma} \end{pmatrix} \begin{pmatrix} \vec{\Gamma} \\ \text{bar}\vec{\Gamma} \end{pmatrix} \right]. \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &\sum_{i=0}^{l_1(x)-1} \left\{ \sum_{i'=0}^{l_2(\sigma_B^i(h(x)))-1} f(\sigma_C^{i'}(g(\sigma_B^i(h(x)))))) - \sum_{j'=0}^{k_2(\sigma_B^i(h(x)))-1} f(\sigma_C^{j'}(g(\sigma_B^i(h(x)))))) \right\} \\ &= \sum_{i'=0}^{l_2^{l_1(x)}(h(x))-1} f(\sigma_C^{i'}(gh(x))) - \sum_{j'=0}^{k_2^{l_1(x)}(h(x))-1} f(\sigma_C^{j'}(g(\sigma_B^{l_1(x)}(h(x)))))) \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathcal{F}(F_{n;\underline{k}}(f_{\underline{k}}))(\xi) &:= \int_{\mathbb{R}^{n+1}} M'_{n+1}(\xi; \underline{\eta}) \varphi_{k_1}(\eta_1) \cdots \varphi_{k_{n+1}}(\eta_{n+1}) \\ &\quad \times \widehat{f_{\underline{k}}}(\xi - \eta_1 - \cdots - \eta_{n+1}) \cdot \widehat{h}(\eta_1) \cdot \dots \cdot \widehat{h}(\eta_{n+1}) d\underline{\eta}, \end{aligned}$$